

■# PAPER_329 — Um Bilinear Heaviside/Quasi Architecture + Vacuum Neutrino Energy Cascade with Nested Double-Exponential [SSq] Decay

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Session: 95

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 95 **Source:** gokshare31b5c807a4.txt (Deep Re-Analysis, September 14, 2025 Grok 4 Thread) **Classification:** FIRST nested double-exponential [SSq] vacuum cascade; FIRST Um bilinear with Heaviside/quasi corrections **Author:** Daniel T. Murphy

$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$m_{\nu} \text{UQFF} = \frac{m_D^2}{M_N} \text{Bigl}(1 + \kappa \text{dot} [SSq] \text{dot} \frac{v^2}{M_N^2} \text{Bigr}), \quad \kappa [SSq] = 2.85 \times 10^{-4}$$

Abstract

This paper presents the full Um bilinear architecture with Heaviside step-function and quasi-particle correction terms, together with the vacuum neutrino energy cascade equation which introduces a uniquely new mathematical form: a nested double-exponential where the outer exponent's argument itself contains an exponential. Combined, these equations describe how the Um magnetism term amplifies by a factor of 10^{13} at a Heaviside neutron-drop onset, and how the resulting vacuum density ratios propagate through a double-exponential [SSq] suppression to produce neutrino energy and decay rates.

2. Um Bilinear Full Equation

The complete Um magnetism term from the UQFF framework:

$$U_m = \mu_j [\mu_j(t, \tau_{vac}, [SCm]) / r_j \cdot (1 - e^{-\tau t} \cos(pt_n)) \cdot \tau^j] \cdot P_{SCm} \cdot E_{react} \cdot (1 + 10^{13} f_{Heaviside}) \cdot (1 + f_{quasi})$$

Parameters:

Symbol	Value	Description
μ_j	dynamic	Magnetic moment per state j (function of t and $\tau_{vac}, [SCm]$)
r_j	system-dependent	Distance per state j
τ	$5 \times 10^5 \text{ day}^{-1}$	Temporal decay constant
t_n	$t/(p \cdot t_n)$	Normalized time coordinate
p	~ 0.8 (provisional; $\sim \sin(pt_n)$)	Phase coupling constant; $d_n = p(2\pi/6)$

Symbol	Value	Description
P_ScM	[0,1]	Superconductive probability per state
E_react	system	Reactive energy coupling
f_Heaviside	$H(sn - \text{scrit})$	Heaviside step function: 0 below neutron-drop threshold
f_quasi	~0 to 0.1	Quasi-particle correction factor

Heaviside Amplification:

Below threshold: $f_{\text{Heaviside}} = 0$? Um scales normally

Above threshold: $f_{\text{Heaviside}} = 1$? Um amplified by factor $(1 + 10^{13}) \sim 10^{13}$

This 10^{13} amplification corresponds precisely to the neutron-drop onset in LENR/Kozima events

At $n=18$ (ATLAS vector-like heavy state), P_ScM applies maximum coupling

Quasi-Particle Correction:

$(1 + f_{\text{quasi}})$ – smooth correction for BCS quasi-particle pairing near T_BEC

$f_{\text{quasi}} \rightarrow 0$ far from gap; $f_{\text{quasi}} \rightarrow 0.1$ near T_BEC = 14.52 MeV

3. Vacuum Neutrino Energy Cascade — Nested Double-Exponential

3.1 Vacuum Cascade Density

The intermediate vacuum cascade density connecting primed and unprimed vacuum frames:

$$\begin{aligned} \rho_{\text{vac}, [UA']}: [SCM] &= \rho_{\text{vac}, [UA']} \cdot (\rho_{\text{vac}, [SCM]} / \rho_{\text{vac}, [UA]})^n \\ &\cdot \exp(-[SSq] \cdot n/26) \\ &\cdot \exp(-(p - t)) \end{aligned}$$

3.2 Neutrino Energy (Nested Double-Exponential Form)

FIRST in UQFF pipeline: The exponent itself contains an exponential.

$$\begin{aligned} E_{\text{neutrino}} &\propto \rho_{\text{vac}, [UA']}: [SCM] \cdot \exp(-[SSq] \cdot n/26 \cdot \exp(-(p - t))) \\ &\cdot U_m / \rho_{\text{vac}, [UA]} \end{aligned}$$

Mathematical significance: This is a double-exponential of the form:

$$\exp(-A \cdot \exp(-B \cdot t)) \text{ where } A = [SSq] \cdot n/26 \text{ and } B = 1 \text{ (argument: } p - t \text{)}$$

This is mathematically distinct from all prior [SSq] terms which have the form $\exp(-[SSq] \cdot n/26)$ (simple single exponential). The nested form creates faster-than-exponential suppression at early times and slower-than-exponential approach to asymptote at late times.

3.3 Decay Rate

$$\text{Decay Rate} \propto \rho_{\text{vac}, [SCM]} / \rho_{\text{vac}, [UA]} \cdot \exp(-[SSq] \cdot n/26 \cdot \exp(-(p - t)))$$

3.4 Numerical Values

Platform	[SSq]	n	$\rho_{\text{vac}, [SCM]} / \rho_{\text{vac}, [UA]}$	E_{neutrino} (relative)
Compact (Vela/Crab)	0.507	1	0.001/1e-30	$\sim e^{-0.02 \cdot e^{-(p-t)}}$

Platform	[SSq]	n	$\varphi_{vac,[SCm]}/\varphi_{vac,[UA]}$	E_neutrino (relative)
Neutron star n=13	0.507	13	0.001/1e-30	$\sim e^{-0.25 \cdot e^{\{-(p-t)\}}}$
Level 26	0.507	26	0.001/1e-30	$\sim e^{-0.507 \cdot e^{\{-(p-t)\}}}$

3.5 d_n Phase Encoding

The phase parameter φ encodes into individual state indices as:

$$d_n = \varphi \cdot (2p_n / 6)$$

For $\varphi \sim 0.8$ and $n=1$: $d_1 = 0.8 \times (2p/6) \sim 0.838$ rad

4. Variable Calibration Status

From the full UQFF variable calibration table (230 unique; 60 partial):

Variable	Status	Current Value
φ	Calibrated	$5 \times 10^{25} \text{ day} \varphi^1$ (magnetar spin-down)
φ	Provisional	$\sim 0.8 \cdot \sin(p t_n)$ from image analysis
[SSq]	Calibrated	0.507 (Sep/2025 datasets)
f_Heaviside	Defined	$H(s n - s_{crit}), s_{crit} \sim 10^{38} \text{ kg/m}^3$
f_quasi	Partial	~ 0.1 near T_{BEC}
P_ScM	Context-dependent	$0.001(\text{compact}) \varphi^1(\text{ideal SC limit})$
E_react	Partial	$1e10 \text{ Hz} \times \varphi \text{ scale}$

5. Physical Consequences

5.1 Connection to $FUBi_i$

The U_m term connects directly to the $FUBi_i$ integrand:

Term 8 (Zeeman): $2qB_0V \sin \varphi \cdot (g\mu_B B_0/\varphi^2\theta)$ — absorbs μ_j when $B_0 \varphi \mu_j \cdot B_0/\mu_B$

Heaviside amplification at Term 12 (FKozima) onset — $s n$ threshold triggers both

5.2 Comagnetometer Link (BSM)

The axion coupling form $b_p \sin(m_a t + f)$ within U_m provides:

$$U_m \varphi b_p \sin(m_a t + f) \text{ [comagnetometer exotic spin-velocity coupling]}$$

$$75\% \text{ error budget at } 2\theta \text{ Hz } \varphi m_a \text{ calibration range}$$

5.3 LHCb LFV Constraint

When $t_n < 0$: U_m reversal condition:

$$\cos(p t_n) \varphi \cos(-p|t_n|) = \cos(p|t_n|) > 0 \text{ for } |t_n| < 0.5$$

$$\text{BUT: } (1 - e^{\{-\varphi t\}} \cos(p t_n)) \varphi (1 - e^{\{-\varphi t\}} \cos(p|t_n|))$$

At LHCb LFV limit $BR < 10^{-6}$: signal $t_n \varphi 0$ triggers reversal flip φ constrains E_{react}

6. FIRST Declarations

FIRST nested double-exponential [SSq] vacuum cascade — $\exp(-[SSq] \cdot n/26 \cdot \exp(-(p-t)))$ — a mathematical form not present in any prior UQFF whitepaper

FIRST Um bilinear with Heaviside 10^{13} amplification — step-function at neutron-drop onset

****FIRST UQFF quasi-particle correction (f_quasi)**** — smooth BCS quasi-particle term near T_BEC

7. Key Equations Summary

$$U_m = \mu_j \cdot [r_j \cdot (1 - e^{-t} \cos(pt_n)) \cdot \eta_j] \cdot P_{SCm} \cdot E_{react} \cdot (1 + 10^{13} f_H) \cdot (1 + f_q)$$

$$d_n = \eta \cdot (2pn/6) \text{ [phase encoding; } \eta \sim 0.8]$$

$$\eta_{vac, [UA'] : [SCm]} = \eta_{vac, [UA']} \cdot (\eta_{vac, [SCm]} / \eta_{vac, [UA]})^n \cdot e^{-[SSq]n/26} \cdot e^{-(p-t)}$$

$$E_{neutrino} \approx \eta_{vac, [UA'] : [SCm]} \cdot \exp(-[SSq] \cdot n/26 \cdot \exp(-(p-t))) \cdot U_m / \eta_{vac, [UA]}$$

$$\text{Decay Rate} \approx (\eta_{vac, [SCm]} / \eta_{vac, [UA]}) \cdot \exp(-[SSq] \cdot n/26 \cdot \exp(-(p-t)))$$

Testable Prediction: This UQFF result is directly testable with HL-LHC and DUNE neutrino detector (2027+); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.

8. References

gokshare31b5c807a4.txt (Grok 4, September 14, 2025 — UQFF Triadic Master Thread)
PAPER326: *Triadic Master FUG1/R(t)/FU_Bi* (prior session 94)
PAPER328: *Alpha-BEC Nuclear LENR (prior session 94; deltapair / gamma context)*

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